Introducing HeGram

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Outline

1. Overview
2. Argument structure in HeGram
3. Implementation
4. Conclusions & future work

HeGram in a nutshell

- A deep linguistic processing grammar of Modern Hebrew.
- Implemented in the LKB (Copestake, 2002) and ACE.
- Developed in parallel with AraGram, a grammar of Modern Standard Arabic.
- Morphology is handled outside the grammar.

(Herzig Sheinfux et al., 2015)

Phenomena covered by HeGram

- Case marking
- Subject-verb and noun-adjective agreement
- Topicalization and wh-questions
- Multiple subcategorization frames
- Control and raising verbs
- Relatively free complement order
- SVO and V2
- The copular construction and zero copula constructions

HeGram & AraGram in numbers

The distribution of types in HeGram & AraGram

	AraGram	HeGram
Number of types	1,888	1,817
Shared (%)	95.5	99.2
Language-specific (%)	4.5	0.8
:+ types (% of shared)	0.3	1.3

Distinguishing between semantic selection and syntactic realization

- A set of semantic categories
- A set of syntactic categories
- For each semantic category - a syntactic realization class
- For each lexeme - a semantic frame and a corresponding syntactic frame
- Constraints can be stated for each level separately

Some existing semantic rolesets

- **Fillmore's Deep Cases & FrameNet**

  (Fillmore, 1968) & (Fillmore et al., 2003)

- **VerbNet**

  (Schuler, 2005)

- **Syntactic Approximation of Semantic Roles for Polish**

  (Jaworski and Przepiórkowski, 2014)

- **PropBank**

  (Palmer et al., 2005) & (Bonial et al., 2012)

- **Handwritten HPSG and DELPH-IN style HPSG**

  (Pollard and Sag, 1994) & (Copestake, 2009)

Defining a roleset

- The semantic categories should be meaningful and consistent across predicates
- The set of roles should be small and manageable
- One argument per semantic category/role

The goal is to strike the right balance between elaborate semantic representations and realistic grammar engineering.

The schema

Label	Semantic Arguments	Syntactic Realization
Arg1	Actor, Perceiver, Causer	NP, PP
Arg2	Theme	NP, VP_{inf}, CP, PP
Arg3	Affectee, Benefactive, Malfactive , Recipient	NP, PP
Arg4	Attribute	AdjP, AdvP, PP, NP, $VP_{beinoni}$
Arg5	Source	PP
Arg6	Goal	PP
Arg7	Location	PP, AdvP
Arg8	Topic of Communication	PP
Arg9	Instrument	PP
Arg10	Comitative	PP

Similar in spirit to work done on Polish by Przepiórkowski et al. (2014).

HeGram's representation of semantic relations

'Standard' HPSG

$$\begin{bmatrix} \textit{take-out-rel} & \\ \textsc{arg}1 & \textit{index} \\ \textsc{arg}2 & \textit{index} \\ \textsc{arg}3 & \textit{index} \end{bmatrix}$$

$$\begin{bmatrix} \textit{come-rel} & \\ \textsc{arg}1 & \textit{index} \\ \textsc{arg}2 & \textit{index} \\ \textsc{arg}3 & \textit{index} \end{bmatrix}$$

HeGram

$$\begin{bmatrix} \textit{take-out-rel} & \\ \textsc{arg}1 & \textit{index} \\ \textsc{arg}2 & \textit{index} \\ \textsc{arg}5 & \textit{index} \end{bmatrix}$$

$$\begin{bmatrix} \textit{come-rel} & \\ \textsc{arg}1 & \textit{index} \\ \textsc{arg}5 & \textit{index} \\ \textsc{arg}6 & \textit{index} \end{bmatrix}$$

Arg1 = Actor; Arg2 = Theme; Arg5 = Source; Arg6 = Goal

Semantic frames & syntactic realizations

(1) a. [dan $_{Arg1}$] amar [et ha-'emet $_{Arg2}$] [la-deriktoriyon $_{Arg3}$] ['al
Dan told ACC the-truth to.the-board about
macav ha-x̲evra$_{Arg8}$]
status the-company

'Dan told the board the truth about the status of the company.'

b. [dan $_{Arg1}$] amar [li $_{Arg3}$] ['al ha-ma'amar$_{Arg8}$] [še-hu m'eod
Dan told to.me about the-article that-it very
x̲adšani $_{Arg2}$]
innovative

'Dan said to me about the article that it is very innovative.'

Semantic frames & syntactic realizations

(2) a. [dan $_{Arg1}$] amar [et ha-'emet $_{Arg2}$] [la-deriktoriyon $_{Arg3}$] ['al
Dan told ACC the-truth to.the-board about
macav ha-xevra$_{Arg8}$]
status the-company
'Dan told the board the truth about the status of the company.'

b. [dan $_{Arg1}$] amar [li $_{Arg3}$] ['al ha-ma'amar$_{Arg8}$] [še-hu m'eod
Dan told to.me about the-article that-it very
xadšani $_{Arg2}$]
innovative
'Dan said to me about the article that it is very innovative.'

Semantic Frame	Syntactic Frame
arg1238	Arg1 ∈ {NP}, Arg2 ∈ {NP,CP}, Arg3 ∈ {PP}, Arg8 ∈ {PP}

Arg1 = Actor; Arg2 = Theme; Arg3 = Recipient; Arg8 = Topic of communication

Advantages

Overview

- Representing verb senses
- Accounting for argument optionality
- Accounting for co-occurrence restrictions
- Generalizing syntactic realization classes
- Capturing the case marking patterns of NPs
- Distinguishing between argument markers and semantic PPs
- Controlling complement order

Representing verb senses

The Saying sense of *amar* ('tell'):

(3) [dan $_{Arg1}$] amar [et ha-'emet $_{Arg2}$] [la-deriktoriyon $_{Arg3}$]
Dan told ACC the-truth to.the-board
'Dan told the board the truth.'

$$\text{Arg2} \in \{\text{NP,CP}\}$$

The Request sense of *amar* ('tell'):

(4) [dan $_{Arg1}$] amar [le-kulam $_{Arg3}$] [lacet me-ha-xeder $_{Arg2}$]
Dan told to-everyone to.exit from-the-room
'Dan told everyone to get out of the room.'

$$\text{Arg2} \in \{\text{VP}\}$$

Argument optionality

(5) a. [dan $_{Arg1}$] amar [et ha-'emet $_{Arg2}$] ([la-deriktoriyon $_{Arg3}$]) (['al macav ha-x̲evra$_{Arg8}$])
Dan told ACC the-truth to.the-board about status the-company

'Dan told the board the truth about the status of the company.'

b. [dan $_{Arg1}$] amar ([li $_{Arg3}$]) (['al ha-ma'amar$_{Arg8}$]) [še-hu m'eod x̲adšani $_{Arg2}$]
Dan told to.me about the-article that-it very innovative

'Dan said to me about the article that it is very innovative.'

Semantic Frame	Realization Frames
arg1238	arg12-123-128-1238

Co-occurrence restrictions

Not all semantic arguments can co-occur:

(6) a. dan x̲ašaš [mi-hefsed $_{Arg2}$] / [lehafsid $_{Arg2}$] / [še-hu yafsid $_{Arg2}$]
Dan feared from-loss / to.lose / that-he will.lose
'Dan feared a loss / to lose / that he will lose.'

b. dan x̲ašaš [le-ʿatid-o $_{Arg3}$]
Dan feared to-future-his
'Dan feared for his future.'

c. *dan x̲ašaš [le-ʿatid-o $_{Arg3}$] [mi-hefsed $_{Arg2}$]
Dan feared to-future-his from-loss

Semantic Frame	Realization Frames
arg123	arg12-13

Generalizing syntactic realization classes

Predicates which semantically select a specific semantic argument (e.g., Arg4) lexically specify which phrase types within the realization class can realize them.

(7) a. dan haya [aluf *NP*] / [me'ušar *AdjP*] / [ba-misxaq *PP*] / [šam *AdvP*]
Dan was champion / happy / in.the-game / there
'Dan was a champion / happy / in the game / there.'

b. dan hiš'ir et ha-ʿinyan [patuax *AdjP*] / [ba-'avir *PP*]
Dan left ACC the-issue open / in.the-air
'Dan left the issue open / in the air.'

c. hi ʿasta et dan [adam ṭov yoter *NP*] / [me'ušar *AdjP*]
she made ACC Dan man good more / happy
'She made Dan a better man / happy.'

d. dan nir'e [zva'a *NP*] / [me'ušar *AdjP*] / [kmo aluf *PP*]
Dan looks horror / happy / like champion
'Dan looks horrible / happy / like a champion.'

Capturing the case marking patterns of NPs

Definite NPs which realize Arg2 and Arg3 complements are marked with accusative case.

(8) a. [dan $_{Arg1}$] axal [maraq $_{Arg2}$]
Dan ate soup
'Dan ate soup.'

b. [dan $_{Arg1}$] axal [et ha-maraq $_{Arg2}$]
Dan ate ACC the-soup
'Dan ate the soup.'

(9) [dan $_{Arg1}$] he'exil [et dana $_{Arg3}$] [maraq $_{Arg2}$]
Dan fed ACC Dana soup
'Dan fed Dana soup.'

Definite NPs which realize Arg4 complements are not marked.

(10) [dan $_{Arg1}$] haya [aluf ha-ʿolam $_{Arg4}$]
Dan was champion the-world
'Dan was the world champion.'

Distinguishing between argument markers and semantic PPs

PPs which realize Arg1, Arg2, and Arg3 are argument-marking PPs.

(11) [dan $_{Arg1}$] ṭipel [ba-tinoq $_{Arg2}$]
Dan looked-after in.the-baby
'Dan looked after the baby.'

PPs which realize Arg4–Arg10 are semantic PPs.

(12) [dan $_{Arg1}$] sam [et ha-sefer $_{Arg2}$] [ba-argaz $_{Arg7}$] / [ʿal ha-argaz $_{Arg7}$]
Dan put ACC the-book in.the-box / on the.box
'Dan put the book in/on the box.'

Controlling complement order

(13) a. [dan $_{Arg1}$] natan [le-dana $_{Arg3}$] [et ha-sefer $_{Arg2}$]
dan gave to-dana ACC book

b. [dan $_{Arg1}$] natan [et ha-sefer $_{Arg2}$] [le-dana $_{Arg3}$]
dan gave ACC book to-dana

'Dan gave the book to Dana.'

Arg2 $\prec$ Arg3, Arg3 $\prec$ Arg2

(14) a. dana ʿasta [et dan $_{Arg3}$] [adam ṭov $_{Arg4}$] / [me'ušar $_{Arg4}$]
Dana made ACC Dan man good / happy

'Dana made Dan a good man/happy.'

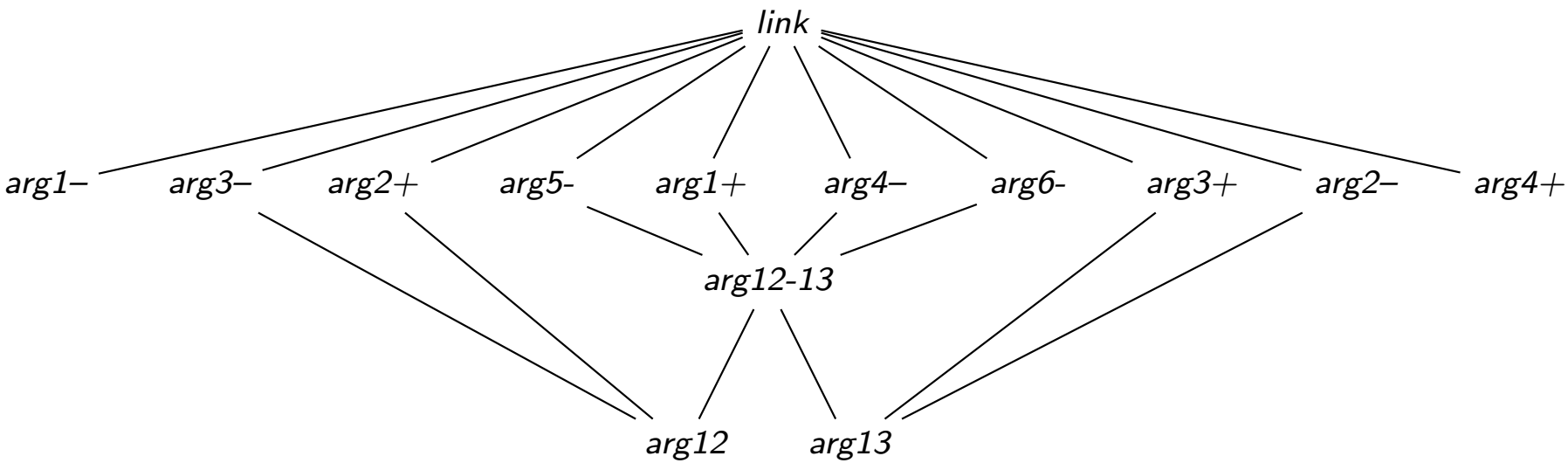
b. *dana ʿasta [adam ṭov $_{Arg4}$] / [me'ušar $_{Arg4}$] [et dan $_{Arg3}$]
Dana made man good / happy ACC Dan

Arg3 $\prec$ Arg4, Arg4 $\not\prec$ Arg3

Overview

- Implemented in the LKB system, based on a Matrix-induced grammar
- Incorporates Haugereid's packed argument frames (Haugereid, 2012)

Haugereid's link hierarchy



(Haugereid, 2012)

VALENCE in HeGram

$$\begin{bmatrix}
\text{R-FRAME} & \textit{link} \\
\text{DEP1} & \begin{bmatrix} \textit{dependent} \\ \text{REAL} \quad \textit{link} \\ \text{DEP} \quad \textit{synsem} \end{bmatrix} \\
\text{DEP2} & \textit{dependent} \\
\ldots & \\
\text{DEP10} & \textit{dependent} \\
\text{PPSORT} & \begin{bmatrix} \text{DEP1-P} \quad \textit{prep-p} \\ \text{DEP2-P} \quad \textit{prep-p} \\ \ldots \\ \text{DEP10-P} \quad \textit{prep-p} \end{bmatrix}
\end{bmatrix}$$

The restructuring of the VALENCE complex is inspired by Haugereid's packed argument frames (Haugereid, 2012).

Example

(15) a. [dan $_{Arg1}$] amar [et ha-'emet $_{Arg2}$] [la-deriktoriyon $_{Arg3}$] ['al
Dan told ACC the-truth to.the-board about
macav ha-$\underline{x}$evra$_{Arg8}$]
status the-company

'Dan told the board the truth about the status of the company.'

b. [dan $_{Arg1}$] amar [li $_{Arg3}$] ['al ha-ma'amar$_{Arg8}$] [še-hu m'eod
Dan told to.me about the-article that-it very
$\underline{x}$adšani $_{Arg2}$]
innovative

'Dan said to me about the article that it is very innovative.'

The lexical level

Lexical Type: Semantic selection and syntactic selection

(16) arg12-123-128-1238_njp_p_le := arg1_np & arg2_np-cp & arg3_pp & arg8_pp &

$$\left[\text{SYNSEM} \mid \text{LOCAL} \mid \text{CAT} \mid \text{VAL} \mid \text{R-FRAME} \quad \textit{arg12-123-128-1238}\right]$$

Lexical Entry: Lexeme-specific information

(17) amar := *arg12-123-128-1238_njp_p_le* &

$$\left[\begin{array}{ll} \text{STEM} & \left\langle \text{“}\textit{amar}\text{”} \;\; (\text{‘}\textit{said}\text{’}) \right\rangle \\ \text{SYNSEM} & \left[\begin{array}{ll} \text{LOCAL} & \left[\begin{array}{ll} \text{CAT} & \left[\begin{array}{ll} \text{HEAD} \mid \text{CNCRD} & \text{PNG-3SM} \\ \text{VAL} \mid \text{PPSORT} & \left[\begin{array}{ll} \text{DEP3-P} & \textit{_l_p_rel} \\ \text{DEP8-P} & \textit{_yl_p_rel} \end{array}\right] \end{array}\right] \end{array}\right] \\ \text{LKEYS} \mid \text{KEYREL} \mid \text{PRED} & \textit{_amar_v_rel} \end{array}\right] \end{array}\right]$$

The valence feature of *amar* ('say')

(18)

$$\begin{bmatrix}
\text{R-FRAME} & \textit{arg12-123-128-1238} \\
\text{S-FRAME} & \textit{arg1238} \\
\text{SUBJ-ARG} & \textit{arg1} \\
\text{DEP1} & \begin{bmatrix}\ldots\text{HEAD} & \textit{noun}\end{bmatrix} \\
\text{DEP2} & \begin{bmatrix}\ldots\text{HEAD} & \textit{+nc}\end{bmatrix} \\
\text{DEP3} & \begin{bmatrix}\ldots\text{HEAD} & \textit{adp}\end{bmatrix} \\
\ldots & \\
\text{DEP8} & \begin{bmatrix}\ldots\text{HEAD} & \textit{adp}\end{bmatrix} \\
\ldots & \\
\text{PPSORT} & \begin{bmatrix}
\ldots & \\
\text{DEP3-P} & \textit{_l_p_rel} \\
\ldots & \\
\text{DEP8-P} & \textit{_yl_p_rel} \\
\ldots &
\end{bmatrix}
\end{bmatrix}$$

The phrasal level

The *head-complement-phrase* schema

(19) head-comp2-phrase := basic-head-init-comp-phrase &

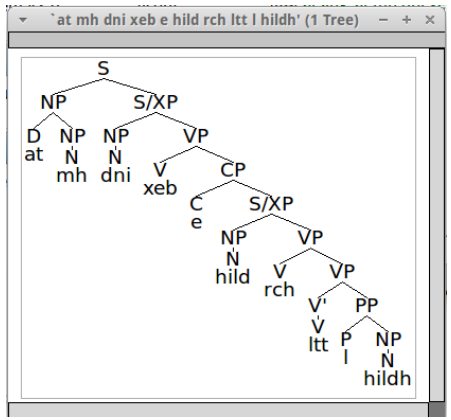
$$\left[\begin{array}{l}\text{SYNSEM}\,|\,\text{LOCAL}\,|\,\text{CAT}\,|\,\text{VAL}\left[\begin{array}{l}\text{R-FRAME}\ \boxed{0}\\ \text{DEP2}\left[\text{REAL}\ \boxed{0}\textit{arg2+}\right]\end{array}\right]\\ \text{HEAD-DTR}\,|\,\text{SYNSEM}\,|\,\text{LOCAL}\,|\,\text{CAT}\,|\,\text{VAL}\left[\begin{array}{l}\text{R-FRAME}\ \boxed{0}\\ \text{SUBJ-ARG}\ \boxed{1}\\ \text{DEP2}\left[\begin{array}{ll}\text{REAL} & \boxed{1}\textit{arg2-}\\ \text{DEP} & \boxed{2}\end{array}\right]\end{array}\right]\\ \text{NON-HEAD-DTR}\,|\,\text{SYNSEM}\ \boxed{2}\end{array}\right]$$

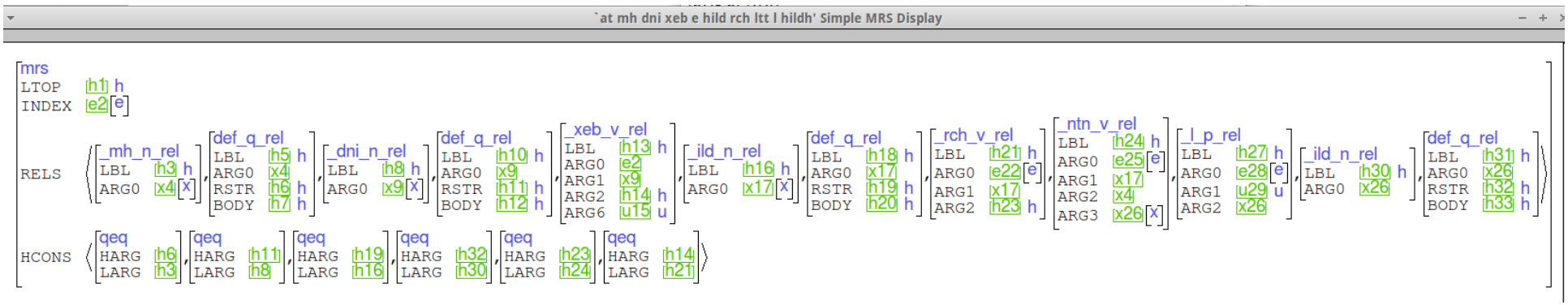
A head can combine with a phrase in a *head-comp2-ph* configuration iff:

- The head subcategorizes for an Arg2.
- Arg2 is not the head's subject.
- The NON-HEAD-DTR is unifiable with its DEP2 requirement.
- The DEP2 requirement of the head hasn't been fulfilled yet.

HeGram in action!

(20) et ma dani x̲ašav še-ha-yeled raca latet la-yalda
ACC what Dan thought that-the-boy wanted to.give to.the-girl
'What did Dan think the boy wanted to give to the girl?'





Conclusions & Future work

Conclusions:

- A solution for the subcategorization problem that meets all the desiderata outlined above
- Striking the right balance between semantic richness and parsimony
- Expressing generalization at the semantic and syntactic level
- It works!

and future work:

- Integration with a wide-coverage lexicon and morphological processors to create a large-scale grammar of Hebrew
- Coverage extension to more phenomena
- Parallel development with AraGram
- Automatic translation between Modern Hebrew and Modern Standard Arabic using generation by MRSs

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